

Prem Krishna Shankar

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EDUCATION

University of Texas at Arlington

Master of Science in Computer Science

Arlington, TX

August 2023 – May 2025

K Ramakrishnan College of Engineering

Bachelor of Engineering in Computer Science

Trichy, India

August 2017 – May 2021

TECHNICAL SKILLS

Languages: C, Java, Python, JavaScript, TypeScript, HTML, CSS

Frameworks & Libraries: React, Node.js, Express.js, Flask, Tailwind CSS, TensorFlow, PyTest

Tools: Docker, Kubernetes, AWS, Azure, Jenkins, Control-M, Git, Selenium, Linux Bash

Databases: PostgreSQL, MongoDB, SQLite

Relevant Courses: Software Engineering, Computer Networks, Machine Learning, Data Structures and Algorithms, Distributed Systems

EXPERIENCE

Assistant Systems Engineer

Tata Consultancy Services

August 2021 – July 2023

Chennai, India

- Automated server maintenance with Bash and Python, reducing manual effort by 40% and downtime by 20%.
- Optimized distributed Control-M workflows orchestrating 10,000+ daily jobs, boosting scheduling efficiency and system uptime from 80% to 90%, ensuring high availability of backend services.
- Engineered Python automation scripts for change request processing, saving 30+ minutes per request.
- Collaborated with cross-functional teams to ensure 100% service availability during power outages, reducing downtime by 50%.
- Led recovery during critical failures, earning “Best Team Award” for exceptional performance in system recovery.

Software Development Intern

Youth Empowerment Foundation

July 2020 – August 2020

Chennai, India

- Developed a full-stack donation platform using JavaScript and Node.js to support COVID-19 relief efforts.
- Implemented a system to collect and manage beneficiary data using Python and MS Excel, helping the organization efficiently track and organize information of 100+ victims.
- Designed digital posters and presentations using HTML, CSS, and Canva to promote campaigns and raise awareness for COVID-19 relief efforts.

PROJECTS

Fotstrike | *HTML, Tailwind CSS, JavaScript, Node.js, Express.js, PostgreSQL, Pytest*

- Built a responsive football web app delivering real-time news, stats, and match data using RapidAPI integration.
- Developed a PostgreSQL-based secure login system and an intuitive frontend UI using HTML and Tailwind CSS to enhance user engagement. The app was tested using Selenium to check the functionality of the website.

Fruit and Vegetable Health Classification | *Python, TensorFlow/Keras, YOLOv8, FastAPI, React*

- Developed a fruit & vegetable health classification system using deep learning (ResNet50, EfficientNetV2B0, MobileNetV2, YOLOv8) achieving 98% test accuracy with EfficientNetV2B0, enabling automated detection of fresh vs. rotten produce.
- Built an interactive FastAPI + React interface for real-time image classification; applied preprocessing (JPEG/RGB conversion, augmentation) on a 28K+ image dataset to improve model robustness and reduce overfitting.

T-Shirt Design Application | *React, Three.js(WebGL), ChatGPT API, Node.js, Docker, Kubernetes*

- Engineered a WebGL-based 3D rendering system using Three.js to enable real-time preview and interaction with T-shirt designs, demonstrating high-performance graphics and 3D modeling capabilities.
- Simulated system-level deployment using Docker and Kubernetes, implementing multi-service architecture to emulate distributed systems and optimize rendering and processing tasks in parallel.